MODEL LAB · 2026-10-15

Every prior, same slate

The held-out scores that decided which prior is served, then how each one would have ranked today's hitters.

Served (full season)
served prior

11
priors compared

26,804
held-out plate appearances

3.05%
held-out HR rate

0.6058
best AUC · KNN comps

Held-out scores

Fitted on plate appearances before 2026-09-03, scored on the 26,804 that came after (actual HR rate 3.05%). No outcome in the test window was visible during fitting. **Read AUC first** — the slate ranks hitters and takes the top six, so ordering is what matters. Brier and log loss are dominated by the ~3% base rate and barely separate the priors. Metrics are per plate appearance; the probabilities lower down are per game.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5778	1.66×	0.02953	0.13565	0.00621
KNN comps Prior is the pooled rate of the nearest comparables.	0.6058	1.55×	0.02947	0.13432	0.00421
SVR Support-vector regression on the same feature matrix.	0.5969	1.53×	0.02952	0.13512	0.00608
Random forest Forest regression; a learned similarity metric.	0.5978	1.53×	0.02951	0.13473	0.00580
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6058	1.55×	0.02947	0.13432	0.00421
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6058	1.55×	0.02947	0.13432	0.00421
Linear regression Ordinary least squares on the same features.	0.6003	1.53×	0.02951	0.13478	0.00444
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6008	1.64×	0.02953	0.13487	0.00658
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6001	1.58×	0.02950	0.13464	0.00483
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.5977	1.55×	0.02954	0.13496	0.00695

BLEND WEIGHTS (FOREST VARIANT)

KNN **1.00** SVR **0.00** RF **0.00**

SHRINKAGE K PER PRIOR

league **257** knn **257** svr **257** rf **257** core **257** ensemble **257** linear **257** logistic **257** nn **257** xgb **257**

Estimated by method of moments on out-of-fold residuals, not hand-tuned.

Thin-sample subset

The same held-out window, restricted to hitters with fewer than 100 plate appearances before the cutoff (3,077 PA, actual HR rate 2.34%). For a regular with 600 PA the observed record swamps any prior, so the priors mostly separate here.

PRIOR	AUC	TOP-DECILE LIFT	BRIER	LOG LOSS	CAL. RMSE
Flat league League HR rate as the prior for every hitter.	0.5874	1.25×	0.02287	0.11127	0.01213
KNN comps Prior is the pooled rate of the nearest comparables.	0.6175	1.94×	0.02281	0.10947	0.01009
SVR Support-vector regression on the same feature matrix.	0.5928	1.94×	0.02307	0.11387	0.01831
Random forest Forest regression; a learned similarity metric.	0.6057	2.36×	0.02302	0.11169	0.01872
Core (KNN+SVR) Blend of KNN and SVR, no forest.	0.6175	1.94×	0.02281	0.10947	0.01009
Forest blend KNN + SVR + forest, weights fitted out-of-fold.	0.6175	1.94×	0.02281	0.10947	0.01009
Linear regression Ordinary least squares on the same features.	0.6136	1.94×	0.02308	0.11210	0.01708
Logistic regression Binomial logistic on the per-PA home run outcome.	0.6205	1.94×	0.02312	0.11132	0.01760
Neural net (torch) Two-layer PyTorch MLP, PA-weighted, trained on the same features.	0.6203	2.22×	0.02286	0.10990	0.01274
XGBoost Gradient-boosted trees, depth 3 with subsampling and L2.	0.6098	1.94×	0.02329	0.11305	0.02431

Slate agreement

PRIOR	SHARED	ITS TOP SIX
Served (full season)	6/6	Jonathan Aranda 15.4%, Junior Caminero 15.0%, Jackson Chourio 15.0%, Shohei Ohtani 14.0%, Jo Adell 13.8%, Victor Mesa Jr. 13.8%
Flat league	5/6	Jackson Chourio 15.1%, Jonathan Aranda 14.6%, Junior Caminero 14.5%, Victor Mesa Jr. 13.6%, William Contreras 13.2%, Jo Adell 13.1%
KNN comps	5/6	Jonathan Aranda 15.6%, Jackson Chourio 15.3%, Junior Caminero 14.9%, Victor Mesa Jr. 14.1%, Shohei Ohtani 14.0%, Jake Bauers 13.7%
SVR	6/6	Junior Caminero 16.2%, Jackson Chourio 15.7%, Jonathan Aranda 15.0%, Victor Mesa Jr. 14.6%, Jo Adell 14.0%, Shohei Ohtani 13.9%
Random forest	5/6	Jackson Chourio 15.8%, Junior Caminero 15.3%, Jonathan Aranda 15.2%, Victor Mesa Jr. 14.3%, Shohei Ohtani 14.2%, Jake Bauers 13.9%
Core (KNN+SVR)	5/6	Jonathan Aranda 15.6%, Jackson Chourio 15.3%, Junior Caminero 14.9%, Victor Mesa Jr. 14.1%, Shohei Ohtani 14.0%, Jake Bauers 13.7%
Forest blend	5/6	Jonathan Aranda 15.6%, Jackson Chourio 15.3%, Junior Caminero 14.9%, Victor Mesa Jr. 14.1%, Shohei Ohtani 14.0%, Jake Bauers 13.7%
Linear regression	5/6	Jackson Chourio 15.5%, Jonathan Aranda 15.3%, Junior Caminero 15.1%, Victor Mesa Jr. 14.2%, Shohei Ohtani 13.9%, Will Smith 13.7%
Logistic regression	6/6	Jackson Chourio 15.6%, Junior Caminero 15.5%, Jonathan Aranda 15.1%, Shohei Ohtani 14.3%, Victor Mesa Jr. 14.1%, Jo Adell 14.1%
Neural net (torch)	6/6	Jackson Chourio 15.6%, Junior Caminero 15.5%, Jonathan Aranda 15.2%, Victor Mesa Jr. 14.3%, Shohei Ohtani 14.2%, Jo Adell 14.0%
XGBoost	5/6	Jackson Chourio 16.1%, Junior Caminero 15.8%, Jonathan Aranda 15.4%, Shohei Ohtani 14.6%, Victor Mesa Jr. 14.0%, Jake Bauers 13.9%

Hitter by hitter

The top 40 bats by the served probability.

	HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
1	Jonathan Aranda TB slate	15.4%	14.6%	15.6%	15.0%	15.2%	15.6%	15.6%	15.3%	15.1%	
2	Junior Caminero TB	15.0%	14.5%	14.9%	16.2%	15.3%	14.9%	14.9%	15.1%	15.5%	
3	Jackson Chourio MIL slate	15.0%	15.1%	15.3%	15.7%	15.8%	15.3%	15.3%	15.5%	15.6%	
4	Shohei Ohtani LAD slate	14.0%	12.6%	14.0%	13.9%	14.2%	14.0%	14.0%	13.9%	14.3%	
5	Jo Adell CLE slate	13.8%	13.1%	13.5%	14.0%	13.6%	13.5%	13.5%	13.7%	14.1%	
6	Victor Mesa Jr. TB	13.8%	13.6%	14.1%	14.6%	14.3%	14.1%	14.1%	14.2%	14.1%	
7	Jake Bauers MIL	13.4%	12.5%	13.7%	13.8%	13.9%	13.7%	13.7%	13.5%	13.5%	
8	William Contreras MIL	13.4%	13.2%	13.4%	13.2%	13.3%	13.4%	13.4%	13.3%	13.3%	
9	Teoscar Hernández LAD	13.4%	12.7%	13.6%	13.5%	13.6%	13.6%	13.6%	13.6%	13.6%	
10	Patrick Bailey CLE	13.4%	12.7%	13.4%	13.2%	12.9%	13.4%	13.4%	13.2%	12.8%	
11	Will Smith LAD	13.2%	12.9%	13.1%	13.1%	13.5%	13.1%	13.1%	13.7%	13.6%	
12	Mookie Betts LAD	12.4%	12.3%	12.2%	12.6%	12.4%	12.2%	12.2%	12.4%	12.4%	
13	Chase DeLauter CLE	12.3%	12.3%	12.2%	12.0%	12.1%	12.2%	12.2%	12.1%	11.9%	
14	Freddie Freeman LAD	12.0%	11.7%	11.8%	11.6%	11.7%	11.8%	11.8%	11.9%	11.7%	
15	Yandy Díaz TB	11.8%	12.2%	11.9%	12.4%	12.2%	11.9%	11.9%	12.0%	11.9%	
16	Brice Turang MIL	11.8%	12.1%	11.7%	11.7%	11.6%	11.7%	11.7%	11.2%	11.1%	
17	Travis Bazzana CLE	11.7%	11.7%	11.5%	11.4%	11.5%	11.5%	11.5%	11.6%	11.3%	
18	Nathaniel Lowe CLE	11.5%	11.3%	11.7%	11.4%	11.7%	11.7%	11.7%	11.5%	11.4%	
19	Christian Yelich MIL	10.7%	10.9%	11.0%	10.6%	10.9%	11.0%	11.0%	11.1%	10.6%	
20	Garrett Mitchell MIL	10.7%	10.2%	10.5%	9.5%	9.8%	10.5%	10.5%	10.0%	9.9%	
21	José Ramírez CLE	10.6%	10.9%	10.7%	10.8%	10.8%	10.7%	10.7%	11.0%	11.2%	
22	Andy Pages LAD	10.0%	9.6%	9.6%	9.4%	9.7%	9.6%	9.6%	9.6%	9.4%	

HITTER	SERVED (FULL SEASON)	FLAT LEAGUE	KNN COMPS	SVR	RANDOM FOREST	CORE (KNN+SVR)	FOREST BLEND	LINEAR REGRESSION	LOGISTIC REGRESSION	NEURAL NET (TO
23 **Petey Halpin** CLE	**8.5%**	9.6%	8.6%	9.0%	8.8%	8.6%	8.6%	9.6%	9.0%	
24 **Kyle Tucker** LAD	**8.2%**	8.3%	8.2%	8.0%	8.1%	8.2%	8.2%	8.1%	7.9%	
25 **Liam Hicks** TB	**7.3%**	8.4%	7.2%	8.1%	7.3%	7.2%	7.2%	7.7%	7.5%	
26 **Brayan Rocchio** CLE	**7.3%**	7.8%	7.2%	7.3%	6.9%	7.2%	7.2%	7.0%	7.0%	
27 **Jonny DeLuca** TB	**7.2%**	7.8%	7.2%	7.1%	6.7%	7.2%	7.2%	7.0%	7.3%	
28 **Richie Palacios** TB	**7.0%**	8.0%	7.1%	6.8%	6.5%	7.1%	7.1%	6.9%	6.9%	
29 **Tommy Edman** LAD	**6.3%**	6.9%	6.2%	6.1%	5.8%	6.2%	6.2%	6.3%	6.2%	
30 **Joey Ortiz** MIL	**5.8%**	7.0%	6.2%	6.3%	6.3%	6.2%	6.2%	6.1%	6.1%	
31 **Cooper Pratt** MIL	**5.5%**	7.2%	5.5%	4.6%	5.2%	5.5%	5.5%	4.7%	5.0%	
32 **Sal Frelick** MIL	**5.4%**	6.9%	5.6%	5.3%	5.3%	5.6%	5.6%	5.4%	5.4%	
33 **Steven Kwan** CLE	**5.3%**	8.7%	5.7%	3.7%	4.3%	5.7%	5.7%	3.7%	5.3%	
34 **Taylor Walls** TB	**4.8%**	6.6%	4.9%	4.1%	4.1%	4.9%	4.9%	5.0%	5.0%	
35 **Chandler Simpson** TB	**2.9%**	6.3%	3.7%	2.3%	1.7%	3.7%	3.7%	1.7%	3.3%	

Per-game probabilities: each prior's per-PA rate run through this hitter's own plate appearances, park, weather, bullpen and times-through-the-order terms. Only the prior changes across the columns — every other term is held fixed, so a left-to-right spread is the prior disagreeing about the hitter, not the context. Shading grows with the probability.

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